Milestone 0: Problem Identification & Project Proposal Report

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1. Proposed Project Title

Elythra: AI-Powered Semantic Resume Screening and Multi-Factor Candidate Ranking System

2. Problem Statement

In modern recruitment, hiring teams receive hundreds to thousands of resumes for every open position. Manually parsing, cross-referencing, and evaluating each applicant against intricate job descriptions is exceptionally labor-intensive, time-consuming, and highly vulnerable to cognitive fatigue and human error.

Furthermore, conventional initial screening workflows rely heavily on rudimentary keyword searches (Ctrl+F) or superficial visual skimming. These approaches fail to capture contextual semantics and frequently introduce unconscious human bias based on candidate names, gender, non-standard formatting aesthetics, or university pedigree. Consequently, highly qualified candidates with non-traditional resume layouts are routinely eliminated, while candidates who artificially 'game' keyword density pass through, resulting in protracted hiring delays, inflated operational recruitment costs, and inequitable hiring decisions.

3. Problem Significance & Motivation

Current Situation:

Recruiters spend an average of only **6 to 10 seconds per resume** during initial evaluations. Faced with high applicant volumes, screening relies on brittle heuristic shortcuts. This manual bottleneck inflates corporate time-to-hire beyond 40 days, elevates cost-per-hire, and leads to significant candidate drop-off.

Why it Needs to be Addressed:

A scalable, transparent, and context-aware decision-support pipeline is vital to modernize recruitment. Replacing rigid keyword filters with semantic Natural Language Processing (NLP) and multi-factor competency matching compresses screening time from weeks to seconds while enforcing objective, merit-based candidate evaluation.

Multi-Stakeholder Impact:

- **Recruiters & HR Teams:** Eliminates screening fatigue, removes repetitive manual review, and accelerates shortlist turnaround by >80%.
- **Job Applicants:** Guarantees fair, standardized evaluation where competency and skills prevail over formatting or unconscious human bias.
- **Organizations & Enterprises:** Reduces operational cost-per-hire, cuts costly vacancy durations, and prevents bad hires.

4. Target Users & Stakeholders

Stakeholder Category	Target Users	Core Value Proposition Delivered
Primary Users	Corporate Recruiters, HR Managers, Talent Acquisition Specialists	Automated batch parsing, instant candidate ranking leaderboard, transparent match breakdown, and 1-click candidate export.
Secondary Users	Job Seekers, University Graduates, Diverse Applicants	Fair, merit-based assessment based strictly on verifiable competency overlap and qualifications rather than demographic factors.
Tertiary Stakeholders	Enterprise Leadership, Department Heads, Staffing Agencies	Accelerated hiring velocity, reduced operational overhead, and auditable recruitment analytics complying with fair-hiring standards.

5. UN Sustainable Development Goals (SDGs) Alignment

UN SDG	Target & Specific Alignment Rationale
SDG 8: Decent Work & Economic Growth Target 8.5	**Promoting Productive Employment & Decent Work:** Streamlines labor market efficiency by matching qualified talent to appropriate job openings rapidly, minimizing labor friction and enhancing economic productivity.
SDG 10: Reduced Inequalities Target 10.2 & 10.3	**Ensuring Equal Opportunity & Eliminating Discrimination:** Anonymizes demographic identifiers and evaluates candidates purely on semantic skill overlap and verified qualifications, actively preventing systemic recruitment bias.

6. Brief Proposed Solution & Technical Architecture

Elythra is an intelligent, automated recruitment decision-support system built with Python, Natural Language Processing (NLP), and Machine Learning. The platform replaces brittle keyword filtering and black-box automation with an explainable, multi-stage architecture comprising four core operational subsystems:

Subsystem Component	Technical Implementation & Functionality
1. Multi-Format Ingestion & Cleaning	Ingests candidate resumes in **PDF, DOCX, and TXT** formats along with unstructured Job Descriptions. Applies text normalisation, regex-based NER, entity stripping (removing contact details to preserve anonymity), and layout parsing.
2. 500+ Categorized Skill Taxonomy	Maintains a rich curated skill knowledge base across 10 specialized technical domains (Programming, Web, Backend, Data Science & AI/ML, Cloud & DevOps, Databases, Testing, Mobile, etc.) capturing both exact and synonymous industry skills.
3. Explainable Multi-Factor Scoring Engine	Computes a composite compatibility score based on four mathematically transparent components: • **Skill Overlap (45%):** Direct and synonym-aware overlap between JD requirements and extracted candidate skills. • **Semantic Text Similarity (35%):** TF-IDF N-gram vectorization and Cosine Similarity measuring contextual relevance. • **Experience Alignment (10%):** Automated candidate work history extraction validated against minimum required tenure. • **Education Verification (10%):** Hierarchical verification of academic degree credentials (B.Tech, M.Tech, MS, BCA/MCA).
4. Interactive Recruiter Intelligence UI	Delivers an executive web dashboard in **Streamlit** featuring an automated candidate leaderboard, visual color-coded skill badges (**Matched Skills** vs. **Missing Skills**), deep-dive Plotly gauge charts, dynamic threshold filtering, and 1-click export to CSV and multi-sheet Excel reports.

7. Expected Outcome & Project Deliverables

✔ **Fully Functional Web Application:** A production-ready Python and Streamlit platform processing multi-resume batches with sub-second latency.

✔ **Explainable Scoring & Transparent Leaderboard:** Clear, quantified candidate rankings accompanied by granular score breakdowns, eliminating black-box opacity.

✔ **Actionable Skill-Gap Diagnostics:** Visual indicators for matched vs. missing skills enabling recruiters to provide objective feedback or identify training requirements.

✔ **Demonstrated Recruitment Velocity Gains:** Over 80% reduction in initial resume screening duration, collapsing screening timelines from weeks to seconds per batch.

✔ **Demographic Bias Mitigation:** Standardized evaluation protocol prioritizing verified competence over formatting variations or demographic background.

✔ **Audit-Ready Compliance Export Center:** One-click generation of structured CSV and multi-tab Excel (.xlsx) reports ensuring regulatory traceability.

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